Problem statement: Voice agent for BFSI lead qualification

Overview

Build, train, and demonstrate an outbound voice agent for banking, financial services, and insurance that qualifies leads for lending products. Each engineer or small team must clone their own voice and deliver a production style solution that includes scripts, NLU mapping, a front end dashboard, tests, and evaluation evidence.

Product coverage

Home loan

Personal loan

Unsecured loan

Loan against property and other secured loans

Gold loan

Commercial vehicle loan

Four wheeler loan

Education loan

MSME business loan

Credit card acquisition

Key requirements

1. Voice cloning and consent

- Record and supply your own voice sample and provide written consent for its use. Do not use anyone else's voice.

2. Privacy and compliance

- Do not use real customer identifiers in demos or logs. Mask or use clearly fictitious PAN numbers, Aadhaar numbers, card numbers, CVV, and OTPs. Describe data retention and masking practices.

3. Pronunciation and jargon focus

- Include and be scored on correct pronunciation of these terms as used in India: PAN card, OTP, PIN, CVV, Aadhaar card.

4. Language support

- English is required. Optionally add Hindi or code mixed Hindi English. Mark the utterances that switch language.

5. Platform disclosure

- List the platforms, model versions, and key configuration choices used for ASR, TTS, NLU, and orchestration.

Functional expectations

- Lead qualification goal

  The agent must identify interest, gather eligibility indicators, and confirm a next step such as a site visit, branch visit, or human callback.

- Scripts and pauses

  Provide full scripts for each product: opening, identity confirmation, qualifying questions, objection handling, consent capture, next step, and close. Mark explicit pause points where the agent expects user input.

- Intent and slot mapping

  Deliver an intent taxonomy, slot definitions, sample utterances, required versus optional slots, and normalization rules (for example salary and loan amount formats).

- Fallback, escalation, and security

  Define clarification prompts, rules for transferring to a human, DTMF or secure handling for sensitive inputs, and backoff logic after repeated failures.

- Logging and metrics

  Provide a logging schema for call metadata, slot status, confidence scores, and latency. Ensure logs mask sensitive data.

- Front end dashboard

  Build a dashboard where stakeholders can view calls with these fields at minimum: product, call timestamp, status (picked, not picked, busy), transcript, call duration, qualification outcome, confidence scores, and a way to replay masked audio snippets. Include filters and basic analytics such as calls by product, success rate, and average handling time.

- Test harness

  Provide at least ten scripted test scenarios per product and automated tests that validate core flows and NLU behavior.

Deliverables (single repository or archive, with README)

1. Source code and deployment instructions, including Dockerfile or platform configs.
2. Cloned voice artifacts or links and synthesis configuration.
3. Final scripts per product annotated with pause points.
4. Intent and slot mapping files and simple conversation flow diagrams.
5. Front end dashboard code or hosted demo link and instructions to run.
6. Test scenarios and automated test harness.
7. Demo recordings: at least three representative calls per product using the cloned voice, including one successful qualification and one failure path; mask all sensitive data.
8. Evaluation report documenting subjective listening scores and objective keyword checks for BFSI jargon, plus test logs.
9. Voice consent form and a short note on data handling and third party licensing.

Evaluation rubric (100 points)

- Voice naturalness and quality: 30 points. Provide blind listening MOS from at least five reviewers.
- Pronunciation accuracy for BFSI jargon: 25 points. Use human validation and ASR keyword hit rates.
- Script quality and user experience: 20 points. Clarity, brevity, consent capture, and qualification effectiveness.
- NLU and slot mapping correctness: 15 points. Measured by test harness pass rate.

- Compliance and privacy handling: 10 points. Masking, consent, and data policy.

Passing threshold: 60 points overall and no category may score zero.

Constraints and rules

- Do not clone any voice other than your own recorded.
- Do not include unmasked real customer identifiers. Simulate OTP and secure inputs and clearly label them as simulated.
- Never request or store full CVV, full card numbers, or full Aadhaar numbers in plain text during demos.
- Document all third party services and licensing implications.

Optional extra credit

- Real time sentiment detection to adapt tone and pace.
- Seamless English Hindi switching.
- Online TTS generation latency under 300 milliseconds.
- SSML driven prosody control for dynamic emphasis.

Submission checklist

Repository or archive with README that explains reproduction steps and maps each artifact to the rubric evidence: source code, voice consent, cloned voice artifacts, scripts, intent and slot files, dashboard demo, test harness and logs, demo recordings (masked), and evaluation report.

Keep the README concise and provide exact commands to run the demo locally. Good luck.